Day 18 - Lecture 18.4: Important Plot Methods



🌐 **Language / भाषा: English** | हिंदी / Hinglish | ⬅️ Back to Lecture 18 Module

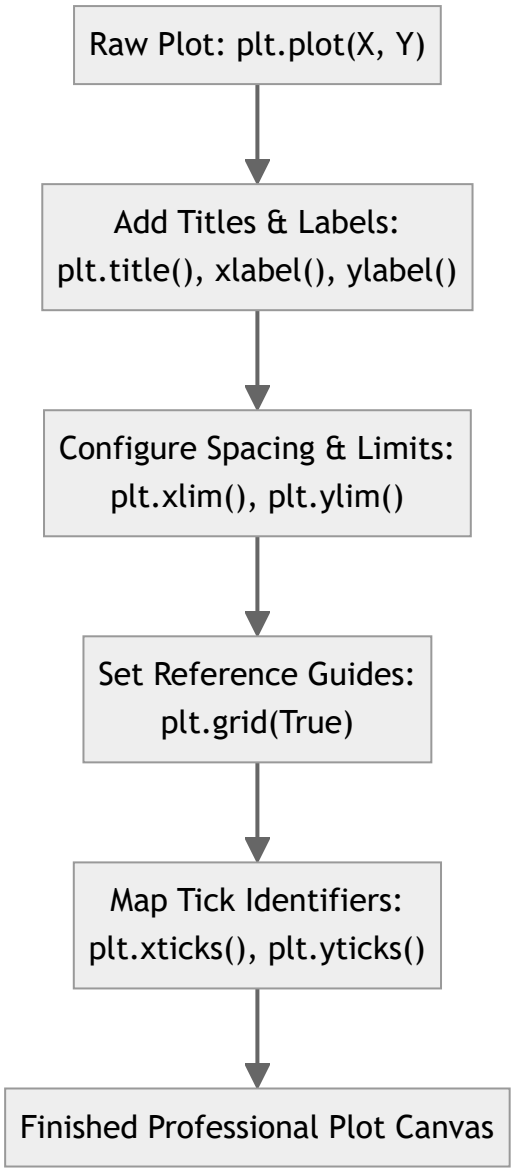
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1. Overview & Conceptual Foundations

Master the primary labeling, scaling, and annotation methods in Matplotlib: title, xlabel, ylabel, xlim, ylim, grid, xticks, and yticks. Proper labeling transforms abstract numbers into self-documenting analytical visualizations.

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2. Visual Architecture & Workflow Diagram



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3. Mathematical & Theoretical Framework

Axis limits define the rectangular viewport subset of $\mathbb{R}^2$:

$$V = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}]$$

Any data coordinate $(x_i, y_i) \notin V$ is clipped by the rasterizer:

$$\text{visible}(x_i, y_i) = \begin{cases} 1 & \text{if } x_{\min} \leq x_i \leq x_{\max} \wedge y_{\min} \leq y_i \leq y_{\max} \\ 0 & \text{otherwise} \end{cases}$$

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4. Structured Implementation & Execution

```
import matplotlib.pyplot as plt

def create_fully_labeled_plot(years: list, revenue: list, movie_title: str) -> dic
    """
    Builds a fully labeled, scaled, and gridded plot adhering to professional
    data visualization standards.

    Parameters:
    -----------
    years : list of int
        Chronological year indices.
    revenue : list of float
        Revenue generated in Millions USD.
    movie_title : str
        Name of the franchise or focus movie category.

    Returns:
    --------
    dict
        Applied axis configuration properties.
    """
    fig, ax = plt.subplots(figsize=(7, 4.5))

    # Plot line with custom markers
    ax.plot(years, revenue, marker='o', markersize=7, color='#7570b3', linewidth=2

    # 1. Labels and Titles
    ax.set_title(f"{movie_title}: Annual Financial Performance", fontsize=13, font
    ax.set_xlabel("Release Year", fontsize=11, fontweight='semibold')
    ax.set_ylabel("Box Office Gross ($ Millions)", fontsize=11, fontweight='semibo

    # 2. Axis limits with padding
    y_padding = max(revenue) * 0.15
    ax.set_ylim(0, max(revenue) + y_padding)
    ax.set_xlim(min(years) - 0.5, max(years) + 0.5)

    # 3. Ticks and Grids
    ax.set_xticks(years)
    ax.grid(True, which='both', linestyle='--', linewidth=0.7, alpha=0.6)
    ax.legend(loc="upper left")

    plt.tight_layout()
    plt.show()

    return {
        "x_ticks": years,
        "y_limits": (0, max(revenue) + y_padding),
        "total_revenue": sum(revenue)
    }
```

```
# Driver execution
years_data = [2008, 2009, 2010, 2011, 2012]
revenue_data = [1005, 170, 427, 133, 232]

config_res = create_fully_labeled_plot(years_data, revenue_data, "Oscar Winners Fr
print("Applied Plot Configuration:")
for k, v in config_res.items():
    print(f"  {k}: {v}")
```

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5. Summary & Key Takeaways

- **Self-contained Functional Design**: Implementation is encapsulated into modular functions with explicit inputs, structured outputs, and type hints.
- **Visual Accuracy**: Adheres strictly to Matplotlib's layout management, aesthetic guidelines, and production best practices.